

Weekly training log, week 9

Week 9

Module	EEEM068, Applied Machine Learning
Project	Industrial waste classification on WaRP
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Focus	EfficientViT B1 on WaRP-C

1. Experimental settings

Dataset	WaRP-C, 8805 train, 1583 test
GPU	NVIDIA RTX 4000 Ada, 20 GB
Input size	224 x 224
Metric	Top 1 accuracy on EMA model
Goal	A small backbone for the efficiency comparison.

2. Hyperparameters

Trial 1

Backbone	EfficientViT B1, ImageNet pretrained
Epochs	60 (10 head only, then 50 full)
Head LR	1e-4
Backbone LR	1e-5
Weight decay	0.05
Optimizer	AdamW with cosine LR
Loss	CE with class weights and LS 0.1 in head phase, then CE with LS 0.1
EMA decay	0.999
Trainable params	around 7 million

3. Results

Trial 1

Epoch	Phase	Loss	Train	Test (EMA)	Best
5	P1	2.166	20.27%	6.51%	6.51%
10	P1	2.021	27.03%	12.31%	12.31%
11	P2	2.168	47.69%	7.87%	12.31%
20	P2	1.244	79.86%	35.65%	35.65%
30	P2	1.057	86.74%	60.48%	60.48%
40	P2	0.967	90.82%	69.83%	69.83%
50	P2	0.932	91.92%	71.76%	72.47%
60	P2	0.932	91.93%	73.31%	73.31%

Best test accuracy: 73.31%. Macro F1: 0.726.

4. Observations

There is a dip at epoch 11 when the backbone unfreezes. EMA takes a couple of epochs to catch up, then it keeps climbing.

73.31% with only 7 million parameters is good for a deployment candidate.